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Electrical connector, method for manufacturing electrical connector, and method for mounting electrical connector

Abstract

An electrical connector includes: a case including an accommodation part that accommodates a receptacle, a front wall part, a restriction part, and an engagement part that retain the receptacle in the accommodation part, a contact part that makes contact with a mounting surface, and a communication hole that is extended upward from the accommodation part and open to outside at its upper end and vertically movably retains a slider; the receptacle that includes a terminal part located at the same height as the contact part or below the contact part and is movable upward; and the slider that can press down the receptacle retained in the accommodation part by the front wall part, the restriction part, and the engagement part.

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Background/Summary

(1) The contents of the following Japanese patent application(s) are incorporated herein by reference: NO. 2022-106384 filed in JP on Jun. 30, 2022 and NO. 2022-193782 filed in JP on Dec. 2, 2022.

FIELD

(2) The present invention relates to an electrical connector including a case and a receptacle to be mounted on a mounting surface of a substrate, a method for manufacturing the electrical connector, and a method for mounting the electrical connector.

BACKGROUND

(3) Electrical connectors configured to retain a receptacle inside a case such as an outer housing have heretofore been known as an electrical connector for in-vehicle use and the like.

(4) Patent Literature 1 discloses an electrical connector including a pair of arm parts that sandwiches a receptacle with an elastic force, and locking parts that each have a locking surface parallel to a plane orthogonal to a direction of insertion of the receptacle and are disposed at the ends of the arm parts. The electrical connector disclosed in Patent Literature 1 also has pressing surfaces at corners formed by the locking surfaces and the arm parts. The pressing surfaces press the receptacle in the insertion direction when the arm parts sandwich the receptacle. The receptacle can thus be stably retained in the outer housing.

(5) The electrical connector disclosed in Patent Literature 1 needs to maintain flatness (hereinafter, referred to as “coplanarity”) of terminal parts of the receptacle with respect to the mounting surface of the substrate. In addition, the electrical connector disclosed in Patent Literature 1 needs to maintain the coplanarity with consideration given to positional accuracy between the fixing position of the receptacle to the case and the connecting position of the case to the substrate. For example, if the receptacle has a terminal pitch of 0.8 mm, a coplanarity of 0.15 mm or less is required.

CITATION LIST

Patent Literature

(6) Patent Literature 1: Japanese Patent Application Laid-Open No. 2016-18720

SUMMARY

Technical Problem

(7) According to Patent Literature 1, the narrower the pitch of the terminal parts of the receptacle is, the more severe the required coplanarity is. Since the positional accuracy between the fixing position of the receptacle to the case and the connecting position of the case to the substrate needs to be taken into consideration, the required coplanarity may be unable to be achieved. For example, in Patent Literature 1, if the receptacle has a terminal pitch of 0.5 mm or less, the coplanarity required of the receptacle is 0.08 mm or less. There has thus been a problem that the required coplanarity is unable to be achieved.

(8) An object of the present invention is to provide an electrical connector that can maintain the coplanarity of the terminal parts of its receptacle favorable even if the terminal parts of the receptacle have a narrow pitch, a method for manufacturing the electrical connector, and a method for mounting the electrical connector.

Solution to Problem

(9) An electrical connector according to an aspect of the present invention is an electrical connector to be mounted on a mounting surface of a substrate, the electrical connector including a case, a receptacle retained in the case, and a slider retained in the case. The case includes an accommodation part that is disposed in a bottom surface to be opposed to the mounting surface and is configured to accommodate the receptacle so that the receptacle is exposed outside, a retaining

part configured to retain the receptacle in the accommodation part, a contact part configured to make contact with the mounting surface, and a communication hole that is extended upward from the accommodation part and open to outside at its upper end, the slider being vertically movably retained in the communication hole. The receptacle includes a terminal part to be soldered to the mounting surface, the terminal part being located at a same height as the contact part or below the contact part and being movable upward in a state where the receptacle is retained in the accommodation part by the retaining part. The slider is capable of moving down in the communication hole to press the receptacle retained in the accommodation part by the retaining part down.

(10) A method for manufacturing an electrical connector according to an aspect of the present invention is a method for manufacturing an electrical connector to be mounted on a mounting surface of a substrate, the method including: an insertion step of inserting a slider into a communication hole and inserting a receptacle into an accommodation part, the accommodation part being disposed in a bottom surface of a case to be opposed to the mounting surface, the communication hole being extended upward from the accommodation part and open to outside at its upper end; and a retaining step of retaining the receptacle inserted into the accommodation part by the insertion step in the accommodation part by a retaining part of the case in a state of being exposed outside so that a terminal part of the receptacle to be soldered to the mounting surface is located at a same height as a contact part of the case to make contact with the mounting surface or below the contact part and the receptacle is movable upward, and retaining the slider inserted into the communication hole by the insertion step in the communication hole in a state of being vertically movable so that the slider moving down in the communication hole is capable of pressing the receptacle retained in the accommodation part by the retaining part down.

(11) A method for mounting an electrical connector according to an aspect of the present invention is a method for mounting an electrical connector to be mounted on a mounting surface of a substrate, the method including, in a state where a receptacle is retained in an accommodation part disposed in a bottom surface of a case to be opposed to the mounting surface by a retaining part of the case in a state of being exposed outside so that a terminal part of the receptacle to be soldered to the mounting surface is located at a same height as a contact part of the case to make contact with the mounting surface or below the contact part and the receptacle is movable upward, and a slider is vertically movably retained in a communication hole that is extended upward from the accommodation part of the case and open to outside at its upper end, transporting the electrical connector to above the mounting surface with the slider sucked to a suction nozzle and bringing the contact part into contact with the mounting surface, pressing the slider down with the suction nozzle to move the slider down so that the slider presses the receptacle down to press the terminal part against the mounting surface, and then soldering the terminal part to the mounting surface.

(12) After the terminal part of the receptacle is brought into contact with the mounting surface of the substrate, the contact part of the case can be brought into contact with the mounting surface while moving the receptacle upward relative to the case. The terminal part of the receptacle is thus soldered to the mounting surface without being affected by the case.

(13) According to the aspect(s) of the present invention, the coplanarity of the terminal parts of the receptacle can be maintained favorable even if the terminal parts of the receptacle have a narrow pitch.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is an exploded perspective view of an electrical connector according to a first embodiment of the present invention.

- (2) FIG. 2 is a perspective view of the electrical connector according to the first embodiment of the present invention, seen from behind and above.
- (3) FIG. 3 is a perspective view of the electrical connector according to the first embodiment of the present invention, seen from behind and below.
- (4) FIG. 4 is a front view of the electrical connector according to the first embodiment of the present invention.
- (5) FIG. 5 is a cross-sectional view taken along the line A-A of FIG. 4.
- (6) FIG. 6 is a perspective view of the electrical connector according to the first embodiment of the present invention with a hold-down removed.
- (7) FIG. 7 is a plan view of the electrical connector according to the first embodiment of the present invention in a state where the electrical connector is mounted on a substrate with the hold-down removed.
- (8) FIG. 8 is a cross-sectional view taken along the line B-B of FIG. 7.
- (9) FIG. 9 is a cross-sectional view of a case of the electrical connector according to the first embodiment of the present invention.
- (10) FIG. 10 is a perspective view of the hold-down of the electrical connector according to the first embodiment of the present invention, seen from rear right and below.
- (11) FIG. 11 is a perspective view of the hold-down of the electrical connector according to the first embodiment of the present invention, seen from rear left and below.
- (12) FIG. 12 is a front view of the electrical connector according to the first embodiment of the present invention in a state where the electrical connector is mounted on the substrate.
- (13) FIG. 13 is a cross-sectional view of the electrical connector according to the first embodiment of the present invention in the state where the electrical connector is mounted on the substrate.
- (14) FIG. 14 is an enlarged cross-sectional view of a part of the electrical connector according to the first embodiment of the present invention before the electrical connector is mounted on the substrate.
- (15) FIG. 15 is an enlarged cross-sectional view of a part of the electrical connector according to the first embodiment of the present invention after the electrical connector is mounted on the substrate.
- (16) FIG. 16 is a perspective view of an electrical connector according to a second embodiment of the present invention.
- (17) FIG. 17 is a perspective view of the electrical connector according to the second embodiment of the present invention with a hold-down removed.
- (18) FIG. 18 is a plan view of the electrical connector including the hold-down according to the second embodiment of the present invention in a state where the electrical connector is mounted on a substrate.
- (19) FIG. 19 is a cross-sectional view taken along line C-C of FIG. 18.
- (20) FIG. 20 is an exploded perspective view of an electrical connector according to a third embodiment of the present invention.
- (21) FIG. 21 is a plan view of the electrical connector including a hold-down according to the third embodiment of the present invention in a state where the electrical connector is mounted on a substrate.
- (22) FIG. 22 is a cross-sectional view taken along line D-D of FIG. 21.

DESCRIPTION OF EMBODIMENTS

(23) Electrical connectors according to embodiments of the present invention will be described in detail below with reference to the drawings as appropriate. In the drawings, an x-axis, a y-axis, and a z-axis constitute a three-axis orthogonal coordinate system. A positive direction on the y-axis will be referred to as forward, a negative direction on the y-axis backward, a positive direction on the x-axis leftward, a negative direction on the x-axis rightward, a positive direction on the z-axis upward, and a negative direction on the z-axis downward.

First Embodiment

(24) <Configuration of Electrical Connector>

(25) A configuration of an electrical connector **1** according to a first embodiment of the present invention will be described in detail below with reference to FIGS. **1** to **14**.

(26) The electrical connector **1** according to the present embodiment includes a case **2**, a hold-down **3**, and a receptacle **4**.

(27) The case **2** is made of an insulating material such as a resin. The case **2** includes a pressing part **18**, a front wall part **19**, projecting wall parts **20**, an accommodation part **21**, engagement parts **22**, contact parts **23**, a mating part **24**, an insertion part **25**, a projecting part **26**, a flat part **27**, a rear wall part **28**, an inner top wall part **29**, and an opening **30**. The front wall part **19**, the projecting wall parts **20**, and the engagement parts **22** constitute a retaining part that restricts forward, backward, leftward, rightward, and downward movement of the receptacle **4** accommodated in the accommodation part **21** to retain the receptacle **4** in the accommodation part **21**.

(28) As shown in FIG. **5**, the pressing part **18** projects below the inner top wall part **29** between projecting parts **412** and terminal parts **422**, to be described below, of the receptacle **4**. The pressing part **18** is disposed opposite to the receptacle **4** retained in the accommodation part **21** with a gap therebetween. The pressing part **18** is designed to make contact with the receptacle **4** when the case **2** and the receptacle **4** are mounted on a mounting surface **100a** of a substrate **100**.

(29) The front wall part **19** is located in front of the accommodation part **21**, and restricts the forward movement of the receptacle **4** retained in the accommodation part **21**.

(30) The projecting wall parts **20** are located behind the engagement parts **22**, and restrict the backward movement of the receptacle **4** accommodated in the accommodation part **21**.

(31) The accommodation part **21** is located at a lower surface or a bottom surface of the case **2** to be opposed to the mounting surface **100a** of the substrate **100**. The accommodation part **21** communicates with outside at its rear end. The accommodation part **21** accommodates the receptacle **4** so that the receptacle **4** is exposed outside.

(32) A pair of engagement parts **22** serving as elastic parts is disposed on the left and right inner walls of the accommodation part **21**. Each engagement part **22** has a locking claw part **221** projecting inward at its lower end. The engagement parts **22** are elastically deformable in a lateral direction between the insertion part **25** and the accommodation part **21**. With the locking claw parts **221** engaged with parts to be engaged **432** of a cover **43**, to be described below, of the receptacle **4**, the engagement parts **22** restrict the downward movement of the receptacle **4**.

(33) The contact parts **23** are located at the lower ends of the left and right side walls of the accommodation part **21** at the front end. The contact parts **23** make contact with the mounting surface **100a** of the substrate **100** when the case **2** is mounted on the mounting surface **100a**.

(34) The mating part **24** is disposed in the projecting part **26**, and has a cylindrical shape open to the outside at the front end. The mating part **24** can be mated with a not-shown counterpart connector.

(35) The insertion part **25** is a penetration hole vertically penetrating the case **2** around the accommodation part **21**. The hold-down **3** is inserted through the insertion part **25**.

(36) The projecting part **26** projects forward from the substrate **100** when the case **2** is mounted on the mounting surface **100a** of the substrate **100** (see FIG. **7**).

(37) The flat part **27** is located at the rear upper surface of the case **2**. The flat part **27** is flat along the front-to-back and left-to-right directions.

(38) The rear wall part **28** is located at the rear end of the case **2**. The rear wall part **28** is recessed forward in the lateral center.

(39) The inner top wall part **29** is located above the accommodation part **21**.

(40) The opening **30** is formed in the front wall part **19**. The accommodation part **21** and the mating part **24** communicate through the opening **30** (see FIG. **5**).

(41) The hold-down **3** is an attachment member formed by punching out a metal plate by press-

working and bending the punched metal plate, and is attached to the case **2** to fix the case **2** to the substrate **100**. The hold-down **3** is inserted through the insertion part **25** of the case **2** and thereby attached to the case **2**. The hold-down **3** includes a top surface part **31**, a rear surface part **32**, internal side surface parts **33**, leg parts **34**, a front surface part **35**, and external side surface parts **36**.

(42) The top surface part **31** covers the flat part **27**.

(43) The rear surface part **32** is formed by bending the metal plate down at the rear end of the top surface part **31**. The rear surface part **32** covers the rear wall part **28**.

(44) The internal side surface parts **33** are formed by bending the metal plate backward at the left and right ends of the front surface part **35**, and located on the left and right of the top surface part **31**. The internal side surface parts **33** are inserted into the insertion part **25**. Each of the internal side surface parts **33** includes a restriction part **331** above and near the front one of a pair of leg parts **34**.

(45) The restriction parts **331** are located laterally outside the locking claw parts **221** and opposed to the locking claw parts **221**. The restriction parts **331** restrict laterally outward elastic deformation of the engagement parts **22** that causes the receptacle **4** to fall out of the accommodation part **21**. Meanwhile, the restriction parts **331** allow laterally outward elastic deformation of the engagement parts **22** so far as the receptacle **4** does not fall out of the accommodation part **21**. Here, laterally outward refers to directions that cause the receptacle **4** to fall out of the accommodation part **21**.

(46) The leg parts **34** serving as soldering parts are extended downward from the respective lower ends of the left and right internal side surface parts **33**. Each internal side surface part **33** has a pair of leg parts **34** spaced in the front-to-back direction. The leg parts **34** project below the insertion part **25** and are located above the bottom surface of the case **2**.

(47) The front surface part **35** is formed by bending the metal plate down at the front end of the top surface part **31**. The front surface part **35** is inserted into the insertion part **25**.

(48) The external side surface parts **36** are formed by bending the metal plate forward at the left and right ends of the rear surface part **32**. The external side surface parts **36** cover the rear end sides of the left and right side surface parts of the case **2**.

(49) The receptacle **4** includes a shield case **41**, contacts **42**, the cover **43**, and a housing **44**.

(50) The shield case **41** is formed by punching out a metal plate by press-working and bending the punched metal plate, and includes a mating part **411** that opens forward and mates with the not-shown counterpart connector. The shield case **41** includes projecting parts **412** that project below the bottom surface. The projecting parts **412** make contact with and are soldered to the mounting surface **100a** of the substrate **100** when the receptacle **4** is mounted on the mounting surface **100a**.

(51) As shown in FIG. 3, a pair of projecting parts **412** is disposed between the terminal parts **422** and the contact parts **23**, on the left and right of a seam of the bent metal plate constituting the shield case **41**. While the projecting parts **412** are soldered to the mounting surface **100a**, the projecting parts **412** are not limited thereto and may simply be put into contact with the mounting surface **100a** without soldering.

(52) The contacts **42** are made of a conductive material and attached to the housing **44**. The contacts **42** include connection parts **421** for connecting to conductive parts of the not-shown counterpart connector, and terminal parts **422** to be soldered to the mounting surface **100a** of the substrate **100**. The terminal parts **422** are located below the contact parts **23** in a state where the locking claw parts **221** of the engagement parts **22** and the parts to be engaged **432** of the cover **43** are engaged to retain the receptacle **4** in the accommodation part **21** (see FIG. 14).

(53) The cover **43** is formed by punching out a metal plate by press-working and bending the punched metal plate, and attached to the shield case **41** as insulated from the contacts **42**. The cover **43** includes leg parts **431** to be connected to not-shown conductive parts of the mounting surface **100a** of the substrate **100**, and the parts to be engaged **432** to be engaged with the locking claw parts **221** of the engagement parts **22**. The parts to be engaged **432** are cut sections which have the

same width as the thickness of the metal plate constituting the cover **43**.

(54) The housing **44** is made of an insulating material such as a resin, and covered with the shield case **41**. The housing **44** retains the contacts **44** so that the contacts **42** are insulated from the shield case **41** and the cover **43**.

(55) When retained in the accommodation part **21**, the receptacle **4** with the foregoing configuration is disposed opposite to the pressing part **18** of the case **2** at a distance and can thus move up relative to the case **2**.

(56) <Method for Manufacturing Electrical Connector>

(57) A method for manufacturing the electrical connector **1** according to the first embodiment of the present invention will be described in detail below with reference to FIGS. **1** to **13**.

(58) Initially, the case **2** is formed by injection molding or the like using a resin material in advance, and the hold-down **3** is formed by press-working a metal plate. Moreover, the receptacle **4** manufactured by an existing method is prepared. Here, a receptacle **4** including terminal parts **422**, the amount or pitch of which corresponds to the use purpose, can be selected.

(59) Next, an insertion step of inserting the receptacle **4** into the accommodation part **21** of the case **2** from in front of and below the projecting wall parts **20** is performed. Here, the receptacle **4** is inserted into the accommodation part **21** while elastically deforming the engagement parts **22** of the case **2** laterally outward with the cover **43** of the receptacle **4**.

(60) Next, a retaining step is performed. In this retaining step, the receptacle **4** is inserted into the accommodation part **21** from below until the parts to be engaged **432** of the cover **43** of the receptacle **4** pass through the locking claw parts **221** of the engagement parts **22**. The locking claw parts **221** of the engagement parts **22** and the parts to be engaged **432** of the cover **43** are thus engaged by the elastic restoring force of the engagement parts **22** of the case **2**, whereby the receptacle **4** is retained in the accommodation part **21**. Here, the receptacle **4** is retained in the accommodation part **21** with its downward movement restricted by the locking claw parts **221** of the engagement parts **22**, its backward movement restricted by the projecting wall parts **20**, and its forward movement restricted by the front wall part **19**.

(61) Next, the leg parts **34** of the hold-down **3** are inserted into the insertion part **25** of the case **2** from above, and the internal side surface parts **33** and the front surface part **35** are then inserted.

(62) The internal side surface parts **33** and the front surface part **35** of the hold-down **3** are then further inserted into the insertion part **25**, whereby the top surface part **31** is brought into contact with the flat part **27** to cover the flat part **27**. The rear surface part **32** covers the rear wall part **28**, and the external side surface parts **36** cover the rear end sides of the left and right side surface parts of the case **2**. The electrical connector **1** is thereby completed.

(63) With the electrical connector **1** completed, the laterally outward elastic deformation of the engagement parts **22** are restricted by the restriction parts **331** of the internal side surface parts **33**. This can prevent the receptacle **4** from falling out of the accommodation part **21** before the electrical connector **1** is mounted on the mounting surface **100a** of the substrate **100**.

(64) Note that the engagement between the locking claw parts **221** and the parts to be engaged **432** is extremely shallow since the parts to be engaged **432** to be engaged with the locking claw parts **221** have an engaging width only as much as the thickness of the metal plate constituting the cover **43**. If the laterally outward elastic deformation of the engagement parts **22** is not restricted by the restriction parts **331**, the locking claw parts **221** and the parts to be engaged **432** are therefore likely to be disengaged and the receptacle **4** to fall out of the case **2** when the electrical connector **1** undergoes an external force such as an impact before the electrical connector **1** is mounted on the substrate **100**.

(65) By contrast, according to the present embodiment, the restriction parts **331** of the hold-down **3** restrict the laterally outward elastic deformation of the engagement parts **22**. This can prevent the receptacle **4** from falling out of the case **2** due to the laterally outward elastic deformation of the engagement parts **22** when the electrical connector **1** undergoes an external force such as an impact,

despite the extremely shallow engagement between the locking claw parts **221** and the parts to be engaged **432**.

(66) <Method for Mounting Electrical Connector>

(67) A method for mounting the electrical connector **1** according to the first embodiment of the present invention will be described in detail below with reference to FIGS. **1** to **15**.

(68) The top surface part **31** of the electrical connector **1** manufactured by the foregoing manufacturing method is sucked to a suction nozzle of a not-shown mounter, and the electrical connector **1** is transported to the mounting surface **100a** of the substrate **100**. Here, as shown in FIG. **14**, the locking claw parts **221** of the engagement parts **22** and the parts to be engaged **432** of the cover **43** are engaged with each other, whereby the terminal parts **422** of the receptacle **4** are located a distance $H1$ below the contact parts **23** of the case **2**. The receptacle **4** is disposed opposite to the pressing part **18** projecting below the inner top wall part **29** of the case **2** at a distance $R1$.

(69) Next, the leg parts **34** of the hold-down **3** of the electrical connector **1** are inserted into through holes **100b** of the substrate **100** from above.

(70) As the leg parts **34** are inserted into the through holes **100b**, the terminal parts **422** of the contacts **42** of the receptacle **4** and the projecting parts **412** of the shield case **41** initially make contact with the mounting surface **100a**. The contact parts **23** of the case **2** then make contact with the mounting surface **100a**.

(71) As shown in FIG. **14**, before the terminal parts **422** and the projecting parts **412** come in contact with the mounting surface **100a**, a vertical distance $H2$ between the lower end parts of the contact parts **23** and the upper end parts of the locking claw parts **221** is smaller than a vertical distance $H3$ between the lower end parts of the parts to be engaged **432** of the receptacle **4** and the soldering surfaces of the terminal parts **422** ($H2 < H3$). The soldering surfaces of the terminal parts **422** and the lower end parts of the projecting parts **412** are located $H1$ below the lower end parts of the contact parts **23**. When the terminal parts **422** and the projecting parts **412** come into contact with the mounting surface **100a** in such a condition, the parts to be engaged **432** move up relative to the locking claw parts **221** and are thereby separated from the locking claw parts **221**. The receptacle **4** moves up relative to the case **2** by the distance $H1$ ($=H3-H2$) inside the accommodation part **21**. This frees the receptacle **4** from the case **2**.

(72) As shown in FIG. **15**, since the receptacle **4** moves up relative to the case **2**, the soldering surfaces of the terminal parts **422** and the lower end parts of the contact parts **23** located at substantially the same positions ($H1 \cong 0$). Since the distance $R1$ is substantially the same as the distance $H1$ ($R1 \cong H1$), the upward movement relative to the case **2** brings the receptacle **4** into contact with the pressing part **18** ($R1=0$), and the receptacle **4** is pressed toward the mounting surface **100a** from above by the pressing part **18**.

(73) Before the electrical connector **1** is mounted on the substrate **100**, the soldering surfaces of the terminal parts **422** of the receptacle **4** are located below the lower end parts of the contact parts **23** of the case **2**. However, this is not restrictive, and the soldering surfaces of the terminal parts **422** and the lower end parts of the contact parts **23** may be vertically located at the same height ($H1=0$ and $H2=H3$). In such a case, the locking claw parts **221** and the parts to be engaged **432** may not be fully separated after contact of the terminal parts **422** and the projecting parts **412** with the substrate **100**. However, the receptacle **4** can be similarly freed from the case **2** if the weight of the receptacle **4** itself does not act on the locking claw parts **221** via the parts to be engaged **432**.

(74) Moreover, the restriction parts **331** of the hold-down **3** allow the laterally outward elastic deformation of the engagement parts **22** so far as the receptacle **4** does not fall out of the accommodation part **21**. The positions of the terminal parts **422** can thus be laterally adjusted to the soldering positions on the mounting surface **100a** for a reliable contact.

(75) Next, the terminal parts **422** and the projecting parts **412** of the receptacle **4** are soldered to not-shown conductive parts of the mounting surface **100a** of the substrate **100**, and the leg parts **34**

of the hold-down **3** are soldered to not-shown conductive parts of the through holes **100b** of the substrate **100**. The mounting of the electrical connector **1** on the substrate **100** is thereby completed. As described above, the parts to be engaged **432** of the receptacle **4** here are disengaged from the locking claw parts **221**. The coplanarity of the terminal parts **422** to the mounting surface **100a** of the substrate **100** can thus be maintained in a favorable state without being affected by the case **2**.

(76) Since the receptacle **4** is pressed against the mounting surface **100a** from above by the pressing part **18**, the terminal parts **422** and the projecting parts **412** can be prevented from being separated from the mounting surface **100a**. Here, the receptacle **4** is pressed against the mounting surface **100a** from above by the pressing part **18** in between the terminal parts **422** in contact with the mounting surface **100a** and the projecting parts **412** in contact with the mounting surface **100a**. The receptacle **4** can thus be mounted in parallel to the mounting surface **100a** without a tilt in the front-to-back direction.

(77) As shown in FIG. 7, the projecting part **26** of the case **2** of the electrical connector **1** mounted on the substrate **100** by the foregoing mounting method projects from the substrate **100**. In such a state, the barycentric position of the electrical connector **1** overlaps the substrate **100** when seen from above (in the state shown in FIG. 7) because of the provision of the hold-down **3**. The electrical connector **1** can thus be stably mounted on the substrate **100**.

(78) The restriction parts **331** are disposed above the leg parts **34** to be soldered to the conductive parts of the through holes **100b**. The laterally outward elastic deformation of the engagement parts **22** can thus be reliably restricted by the restriction parts **331** even if the restriction parts **331** are pressed laterally outward by the engagement parts **22** in the state where the electrical connector **1** is mounted on the mounting surface **100a** of the substrate **100**.

(79) As described above, according to the present embodiment, the electrical connector **1** includes the case **2** and the receptacle **4** to be mounted on the mounting surface **100a** of the substrate **100**. The case **2** includes: the accommodation part **21** that is disposed in the bottom surface to be opposed to the mounting surface **100a** and accommodates the receptacle **4** in a state of being exposed outside; the front wall part **19**, the projecting wall parts **20**, and the engagement parts **22** that retain the receptacle **4** in the accommodation part **21**; and the contact parts **23** that make contact with the mounting surface **100a**. The receptacle **4** includes the terminal parts **422** to be soldered to the mounting surface **100a**. In the state where the receptacle **4** is retained in the accommodation part **21** by the front wall part **19**, the projecting wall parts **20**, and the engagement parts **22**, the terminal parts **422** are located at the same height as the contact parts **23** or below the contact parts **23**, and can move up. The coplanarity of the terminal parts **422** of the receptacle **4** can thus be maintained in a favorable state even if the terminal parts **422** of the receptacle **4** have a narrow pitch.

(80) According to the present embodiment, the electrical connector **1** includes the case **2** to be mounted on the mounting surface **100a** of the substrate **100**, the receptacle **4** to be mounted on the mounting surface **100a** as accommodated in the case **2**, and the hold-down **3** that is attached to the case **2**. The case **2** includes the accommodation part **21** that accommodates the receptacle **4**, and the front wall part **19**, the projecting wall parts **20**, and the engagement parts **22** that retain the receptacle **4** in the accommodation part **21**. The retaining part includes the elastically deformable engagement parts **22** that retain the receptacle **4** in the accommodation part **21** with an elastic force. The hold-down **3** includes the restriction parts **331** that restricts elastic deformation of the engagement parts **22** in the direction of causing the receptacle **4** to fall out of the accommodation part **21**. The receptacle **4** can thereby be prevented from falling out of the case **2**.

(81) According to the present embodiment, the hold-down **3** that fixes the case **2** to the substrate **100** restricts the elastic deformation of the engagement parts **22** in the direction of causing the receptacle **4** to fall out of the accommodation part **21**. The hold-down **3** can thus serve both as a member for fixing the case **2** to the substrate **100** and as a member for restricting the elastic

deformation of the engagement parts **22** in the direction of causing the receptacle **4** to fall out of the accommodation part **21**. This can prevent an increase in the number of components.

(82) In the present embodiment, the elastic deformation of the engagement parts **22** in the direction of causing the receptacle **4** to fall out of the accommodation part **21** is restricted by the hold-down **3**. However, this is not restrictive. An attachment member other than the hold-down **3** may be attached to the case **2**, and the elastic deformation of the engagement parts **22** in the direction of causing the receptacle **4** to fall out of the accommodation part **21** may be restricted by the attachment member.

Second Embodiment

(83) <Configuration of Electrical Connector>

(84) A configuration of an electrical connector **11** according to a second embodiment of the present invention will be described in detail below with reference to FIGS. **16** to **19**.

(85) In FIGS. **16** to **19**, portions having the same configurations as in FIGS. **1** to **15** are denoted by the same reference numerals. A description thereof will be omitted.

(86) The electrical connector **11** according to the present embodiment includes a receptacle **4**, a case **5**, and a hold-down **6**.

(87) The case **5** is made of an insulating material such as a resin. The case **5** includes a front wall part **19**, projecting wall parts **20**, an accommodation part **21**, engagement parts **22**, contact parts **23**, a mating part **24**, an insertion part **25**, a projecting part **26**, an inner top wall part **29**, an opening **30**, spring parts **51**, a movable part **52**, and a rear wall part **53**. The spring parts **51** and the movable part **52** constitute a pressure spring part.

(88) The accommodation part **21** is a recess formed in the lower surface or bottom surface of the case **5**. The accommodation part **21** communicates with outside at the rear end.

(89) The insertion part **25** is a penetration hole vertically penetrating the case **5** around the accommodation part **21**. The hold-down **6** is inserted through the insertion part **25**.

(90) The projecting part **26** projects forward from a substrate **100** when the case **5** is mounted on a mounting surface **100a** of the substrate **100**.

(91) The inner top wall part **29** is located above the accommodation part **21** and opposed to the receptacle **4** retained in the accommodation part **21** at a distance.

(92) The spring parts **51** connect the movable part **52** and the rear wall part **53**. The spring parts **51** are elastically deformed to allow downward movement of the movable part **52** when the movable part **52** is pressed down. The spring parts **52** move the movable part **52** up with its elastic restoring force when the downward pressing force on the movable part **52** is released.

(93) The movable part **52** is located above the accommodation part **21** and opposed to the receptacle **4** retained in the accommodation part **21** at a distance. When pressed down, the movable part **52** moves down to press the receptacle **4** retained in the accommodation part **21** down. When the downward pressing force is released, the movable part **52** moves up to release the downward pressure on the receptacle **4** retained in the accommodation part **21**.

(94) The rear wall part **53** is located at the rear end of the case **5**, on the left and right of the movable part **52**. The rear wall part **53** is recessed forward toward the movable part **52**.

(95) The hold-down **6** is an attachment member formed by punching out a metal plate by press-working and bending the punched metal plate and attached to the case **5** to fix the case **5** to the substrate **100**. The hold-down **6** is inserted into the insertion part **25** of the case **5** and thereby attached to the case **5**. The hold-down **6** includes a rear surface part **32**, internal side surface parts **33**, leg parts **34**, a front surface part **35**, a top surface part **61**, and a window hole part **62**.

(96) The rear surface part **32** is formed by bending the metal plate down at the rear end of the top surface part **61**. The rear surface part **32** covers the rear end of the movable part **52** and the rear wall part **53**.

(97) The internal side surface parts **33** are formed by bending the metal plate backward at the left and right ends of the front surface part **35**, and located on the left and right of the top surface part

61. The internal side surface parts **33** are inserted into the insertion part **25**.

(98) The leg parts **34** project down from the insertion part **25** and are located above the bottom surface of the case **5**.

(99) The front surface part **35** is formed by bending the metal plate down at the front end of the top surface part **61**, and inserted into the insertion part **25**.

(100) The top surface part **61** covers the spring parts **51** and the upper end of the movable part **52**.

(101) The window hole part **62** is a penetration hole that is located substantially in the center of the top surface part **61** and penetrates the top surface part **61** in the thickness direction. The window hole part **62** exposes a part of the movable part **52** outside.

(102) In the electrical connector **11** having the foregoing configuration, laterally outward elastic deformation of the engagement parts **22** is restricted by restriction parts **331** of the internal side surface parts **33**. This can prevent the receptacle **4** from falling out of the accommodation part **21** before the electrical connector **11** is mounted on the substrate **100**.

(103) A method for manufacturing the electrical connector **11** according to the present embodiment is similar to the method for manufacturing the electrical connector **1**. A description thereof will thus be omitted.

(104) <Method for Mounting Electrical Connector>

(105) A method for mounting the electrical connector **11** according to the second embodiment of the present invention will be described in detail below with reference to FIGS. **16** to **19**.

(106) The top surface of the movable part **52** of the case **5** exposed outside from the window hole part **62** of the hold-down **6** of the electrical connector **11** is sucked to a suction nozzle **200** of a mounter, and the electrical connector **11** is transported to the mounting surface **100a** of the substrate **100**. Here, the locking claw parts **221** of the engagement parts **22** and the parts to be engaged **432** of the cover **43** are engaged with each other, whereby the terminal parts **422** of the receptacle **4** are located below the contact parts **23** of the case **5**. The receptacle **4** is disposed opposite to the inner top wall part **29** and the movable part **52** of the case **5** at a distance.

(107) Next, the leg parts **34** of the hold-down **6** of the electrical connector **11** are inserted into the through holes **100b** of the substrate **100** from above.

(108) As the leg parts **34** are inserted into the through holes **100b**, the terminal parts **422** of the contacts **42** of the receptacle **4** and the projecting parts **412** of the shield case **41** initially make contact with the mounting surface **100a**. The contact parts **23** of the case **5** then make contact with the mounting surface **100a**.

(109) Before the contact of the terminal parts **422** and the projecting part **412** with the mounting surface **100a**, the vertical distance between the lower end parts of the contact parts **23** and the upper end parts of the locking claw parts **221** is smaller than the vertical distance between the lower end parts of the parts to be engaged **432** of the receptacle **4** and the soldering surfaces of the terminal parts **422**. The soldering surfaces of the terminal parts **422** and the lower end parts of the projecting parts **412** are located below the lower end parts of the contact parts **23**. When the terminal part **422** and the projecting parts **412** in such a state come into contact with the mounting surface **100a**, the parts to be engaged **432** move up relative to the locking claw parts **221** and are thereby separated from the locking claw parts **221**. Moreover, the receptacle **4** moves up relative to the case **5** inside the accommodation part **21**. This frees the receptacle **4** from the case **5**.

(110) Since the receptacle **4** moves up relative to the case **5**, the terminal parts **422** of the receptacle **4** are vertically located at substantially the same position as the contact parts **23** of the case **5**. The receptacle **4** that moves up relative to the case **5** is disposed opposite to the inner top wall part **29** and the movable part **52** of the case **5** closer than before the mounting on the mounting surface **100a**.

(111) Before the mounting of the electrical connector **11** on the substrate **100**, the soldering surfaces of the terminal parts **422** of the receptacle **4** are located below the lower end parts of the contact parts **23** of the case **5**. However, this is not restrictive, and the soldering surfaces of the

terminal parts **422** and the lower end parts of the contact parts **23** may be vertically located at the same height. In such a case, the locking claw parts **221** and the parts to be engaged **432** may not be fully separated after the contact of the terminal parts **422** with the substrate **100**. However, the receptacle **4** is similarly freed from the case **5** if the own weight of the receptacle **4** does not act on the locking claw parts **221** via the parts to be engaged **432**.

(112) Moreover, the restriction parts **331** of the hold-down **6** allow the laterally outward elastic deformation of the engagement parts **22** so far as the receptacle **4** does not fall out of the accommodation part **21**. The positions of the terminal parts **422** can thus be laterally adjusted to the soldering positions on the mounting surface **100a** for reliable contact.

(113) Next, the movable part **52** of the case **5** is pressed down by the suction nozzle **200**, whereby the spring parts **51** are elastically deformed to move down and press the receptacle **4** down. This presses the terminal parts **422** of the receptacle **4** against not-shown conductive parts of the mounting surface **100a** of the substrate **100**. The terminal parts **422** can thus be stably connected to the conductive parts.

(114) Next, the terminal parts **422** and the projecting parts **412** of the receptacle **4** are soldered to the not-shown conductive parts of the mounting surface **100a** of the substrate **100**, and the leg parts **34** of the hold-down **6** are soldered to not-shown conductive parts of the through holes **100b** of the substrate **100**. The mounting of the electrical connector **11** on the substrate **100** is thereby completed. As described above, the receptacle **4** here is free from the case **5**. The coplanarity of the terminal parts **422** with respect to the mounting surface **100a** of the substrate **100** can thus be maintained in a favorable state without being affected by the case **5**.

(115) As shown in FIG. **18**, the projecting part **26** of the case **5** of the electrical connector **11** mounted on the substrate **100** by the foregoing mounting method projects from the substrate **100**. In such a state, the barycentric position of the electrical connector **11** overlaps the substrate **100** when seen from above (in the state shown in FIG. **18**) because of the provision of the hold-down **6**. The electrical connector **11** can thus be stably mounted on the substrate **100**.

(116) The restriction parts **331** are disposed above the leg parts **34** to be soldered to the conductive parts of the through holes **100b**. The laterally outward elastic deformation of the engagement parts **22** can thus be reliably restricted by the restriction parts **331** in the state where the electrical connector **11** is mounted on the mounting surface **100a** of the substrate **100**.

(117) As described above, according to the present embodiment, the case **5** includes the spring parts **51** and the movable part **52** that are located above the accommodation part **21** and can be elastically deformed to press down the receptacle **4** retained in the accommodation part **21** by the engagement parts **22**. In addition to the effects of the foregoing first embodiment, the receptacle **4** can thus be stably mounted on the mounting surface **100a** of the substrate **100**.

(118) In the present embodiment, in mounting the electrical connector **11** on the mounting surface **100a**, the movable part **52** is pressed down by the suction nozzle **200**. However, this is not restrictive, and the movable part **52** may be pressed down by a jig or the like other than the suction nozzle **200**.

(119) In the present embodiment, the hold-down **6** is configured without the external side surface parts **36**. However, this is not restrictive, and the hold-down **6** may include the external side surface parts **36**.

(120) In the present embodiment, the elastic deformation of the engagement parts **22** in the direction of causing the receptacle **4** to fall out of the accommodation part **21** is restricted by the hold-down **6**. However, this is not restrictive. An attachment member other than the hold-down **6** may be attached to the case **5**, and the elastic deformation of the engagement parts **22** in the direction of causing the receptacle **4** to fall out of the accommodation part **21** may be restricted by the attachment member.

Third Embodiment

(121) <Configuration of Electrical Connector>

(122) A configuration of an electrical connector **12** according to a third embodiment of the present invention will be described in detail below with reference to FIGS. **20** to **22**.

(123) In FIGS. **20** to **22**, portions having the same configurations as in FIGS. **1** to **15** are denoted by the same reference numerals. A description thereof will be omitted.

(124) The electrical connector **12** according to the present embodiment includes a receptacle **4**, a case **7**, a hold-down **8**, and a slider **9**.

(125) The case **7** is made of an insulating material such as a resin. The case **7** includes a front wall part **19**, projecting wall parts **20**, an accommodation part **21**, engagement parts **22**, contact parts **23**, a mating part **24**, an insertion part **25**, a projecting part **26**, a rear wall part **28**, an inner top wall part **29**, an opening **30**, a communication hole **71**, and a flat part **72**.

(126) The accommodation part **21** is a recess formed in the lower surface or bottom surface of the case **7**. The accommodation part **21** communicates with outside at the rear end.

(127) The contact parts **23** make contact with a mounting surface **100a** of a substrate **100** when the case **7** is mounted on the mounting surface **100a**.

(128) The insertion part **25** is a penetration hole vertically penetrating the case **7** around the accommodation part **21**. The hold-down **8** is inserted into the insertion part **25**.

(129) The projecting part **26** projects forward from the substrate **100** when the case **7** is mounted on the mounting surface **100a** of the substrate **100** (see FIG. **18**).

(130) The rear wall part **28** is located at the rear end of the case **7**. The rear wall part **28** is recessed forward in the lateral center.

(131) The inner top wall part **29** is located above the accommodation part **21**. The inner top wall part **29** is disposed opposite to the receptacle **4** retained in the accommodation part **21** at a distance.

(132) The communication hole **71** is extended upward from the accommodation part **21** and open to outside at the upper end. The slider **9** is vertically movably retained in the communication hole **71**.

(133) The flat part **72** is disposed at the rear upper surface of the case **7** around the communication hole **71**. The flat part **72** is flat along the front-to-back and left-to-right directions.

(134) The hold-down **8** is an attachment member formed by punching out a metal plate by press-working and bending the punched metal plate and attached to the case **7** to fix the case **7** to the substrate **100**. The hold-down **8** is inserted into the insertion part **25** of the case **7** and thereby attached to the case **7**. The hold-down **8** includes a rear surface part **32**, internal side surface parts **33**, leg parts **34**, a front surface part **35**, a top surface part **81**, and a window hole part **82**.

(135) The rear surface part **32** is formed by bending the metal plate down at the rear end of the top surface part **81**. The rear surface part **32** covers the rear wall part **28**.

(136) The internal side surface parts **33** are formed by bending the metal plate backward at the left and right ends of the front surface part **35**, and located on the left and right of the top surface part **81**. The internal side surface parts **33** are inserted into the insertion part **25**.

(137) The leg parts **34** project down from the insertion part **25**, and are located above the bottom surface of the case **7**.

(138) The front surface part **35** is formed by bending the metal plate down at the front end of the top surface part **81**. The front surface part **35** is inserted into the insertion part **25**.

(139) The top surface part **81** covers the flat part **72**.

(140) The window hole part **82** is a penetration hole that is located substantially in the center of the top surface part **81** and penetrates the top surface part **81** in the thickness direction. The window hole part **82** exposes the upper end of the slider **9** retained in the communication hole **71** outside.

(141) The slider **9** is cylindrical in shape and vertically movably retained in the communication hole **71**. The slider **9** can press the receptacle **4** down by moving down.

(142) <Method for Manufacturing Electrical Connector>

(143) A method for manufacturing the electrical connector **12** according to the third embodiment of the present invention will be described in detail below with reference to FIGS. **16** to **18**.

(144) Initially, the case **7** is formed by injection molding or the like using a resin material in advance, and the hold-down **8** is formed by press-working a metal plate. Moreover, the receptacle **4** manufactured by an existing method is prepared.

(145) Next, an insertion step of inserting the slider **9** into the communication hole **71** of the case **7** from below and then inserting the receptacle **4** into the accommodation part **21** of the case **7** from in front of and below the projecting wall parts **20** is performed. Here, the receptacle **4** is inserted into the accommodation part **21** while elastically deforming the engagement parts **22** of the case **7** outward by the cover **43** of the receptacle **4**. The rest of the manufacturing method is similar to that of the method for manufacturing the electrical connector **1**. A description thereof will thus be omitted.

(146) With the electrical connector **12** completed, the laterally outward elastic deformation of the engagement parts **22** is restricted by the restriction parts **331** of the internal side surface parts **33**. This can prevent the receptacle **4** from falling out of the accommodation part **21** before the electrical connector **12** is mounted on the substrate **100**.

(147) <Method for Mounting Electrical Connector>

(148) A method for mounting the electrical connector **12** according to the third embodiment of the present invention will be described in detail below with reference to FIGS. **16** to **18**.

(149) The slider **9** exposed outside from the window hole part **82** of the hold-down **8** of the electrical connector **12** manufactured by the foregoing manufacturing method is sucked to a suction nozzle **200** of a mounter, and the electrical connector **12** is transported to the mounting surface **100a** of the substrate **100**. Here, the locking claw parts **221** of the engagement parts **22** and the parts to be engaged **432** of the cover **43** are engaged with each other, whereby the terminal parts **422** of the receptacle **4** are located below the contact parts **23** of the case **7**. The receptacle **4** is disposed opposite to the inner top wall part **29** of the case **7** at a distance.

(150) Next, the leg parts **34** of the hold-down **8** of the electrical connector **12** are inserted into the through holes **100b** of the substrate **100** from above.

(151) As the leg parts **34** are inserted into the through holes **100b**, the terminal parts **422** of the contacts **42** of the receptacle **4** and the projecting parts **412** of the shield case **41** initially make contact with the mounting surface **100a**. The contact parts **23** of the case **7** then make contact with the mounting surface **100a**.

(152) Before the contact of the terminal parts **422** and the projecting parts **412** with the mounting surface **100a**, the vertical distance between the lower end parts of the contact parts **23** and the upper end parts of the locking claw parts **221** is smaller than the vertical distance between the lower end parts of the parts to be engaged **432** of the receptacle **4** and the soldering surfaces of the terminal parts **422**. The soldering surfaces of the terminal parts **422** and the lower end parts of the projecting parts **412** are located below the lower end parts of the contact parts **23**. When the terminal parts **422** and the projecting parts **412** in such a state come into contact with the mounting surface **100a**, the parts to be engaged **432** move up relative to the locking claw parts **221** and are thereby separated from the locking claw parts **221**. Moreover, the receptacle **4** moves up relative to the case **7** inside the accommodation part **21**. This frees the receptacle **4** from the case **7**.

(153) Since the receptacle **4** moves up relative to the case **7**, the terminal parts **422** of the receptacle **4** are vertically located at substantially the same position as the contact parts **23** of the case **7**. The receptacle **4** that moves up relative to the case **7** is disposed opposite to the inner top wall part **29** of the case **7** closer than before the mounting on the mounting surface **100a**, and pushes up the slider **9**.

(154) Before the mounting of the electrical connector **12** on the substrate **100**, the soldering surfaces of the terminal parts **422** of the receptacle **4** are located below the lower end parts of the contact parts **23** of the case **7**. However, this is not restrictive, and the soldering surfaces of the terminal parts **422** and the lower end parts of the contact parts **23** may be vertically located at the same height. In such a case, the locking claw parts **221** and the parts to be engaged **432** may not be

fully separated after the contact of the terminal parts **422** with the substrate **100**. However, the receptacle **4** is similarly freed from the case **7** if the own weight of the receptacle **4** does not act on the locking claw parts **221** via the parts to be engaged **432**.

(155) Moreover, the restriction parts **331** of the hold-down **8** allow the laterally outward elastic deformation of the engagement parts **22** so far as the receptacle **4** does not fall out of the accommodation part **21**. The positions of the terminal parts **422** can thus be laterally adjusted to the soldering positions on the mounting surface **100a** for reliable contact.

(156) Next, the slider **9** is pressed down by the suction nozzle **200** to move down inside the communication hole **71**, and presses the receptacle **4** down. The terminal parts **422** of the receptacle **4** are thereby pressed against not-shown conductive parts of the mounting surface **100a** of the substrate **100**, and can thus be stably connected to the conductive parts.

(157) Next, the terminal parts **422** and the projecting parts **412** of the receptacle **4** are soldered to the not-shown conductive parts of the mounting surface **100a** of the substrate **100**, and the leg parts **34** of the hold-down **8** are soldered to not-shown conductive parts of the through holes **100b** of the substrate **100**. The mounting of the electrical connector **12** on the substrate **100** is thereby completed. As described above, since the receptacle **4** here is free from the case **7**, the coplanarity of the terminal parts **422** with respect to the mounting surface **100a** of the substrate **100** can be maintained in a favorable state without being affected by the case **7**.

(158) As shown in FIG. **21**, the projecting part **26** of the case **7** of the electrical connector **12** mounted on the substrate **100** by the foregoing mounting method projects from the substrate **100**. In such a state, the barycentric position of the electrical connector **12** overlaps the substrate **100** when seen from above (in the state shown in FIG. **21**) because of the provision of the hold-down **8**. The electrical connector **12** can thus be stably mounted on the substrate **100**.

(159) Moreover, since the restriction parts **331** are disposed above the leg parts **34** to be soldered to the conductive parts of the through holes **100b**, the laterally outward elastic deformation of the engagement parts **22** can be reliably restricted by the restriction parts **331** in the state where the electrical connector **12** is mounted on the mounting surface **100a** of the substrate **100**.

(160) As described above, according to the present embodiment, the case **7** includes the communication hole **71** that is extended upward from the accommodation part **21** and open to outside at the upper end, and in which the slider **9** is vertically movably retained. The slider **9** can move down in the communication hole **71** to press down the receptacle **4** retained in the accommodation part **21** by the engagement parts **22**. In addition to the effect of the foregoing first embodiment, the receptacle **4** can thus be stably mounted on the mounting surface **100a** of the substrate **100**.

(161) In the present embodiment, the slider **9** is pressed down by the suction nozzle **200** in mounting the electrical connector **12** on the mounting surface **100a**. However, this is not restrictive, and the slider **9** may be pressed down by a jig or the like other than the suction nozzle **200**.

(162) In the present embodiment, the elastic deformation of the engagement parts **22** in the direction of causing the receptacle **4** to fall out of the accommodation part **21** is restricted by the hold-down **8**. However, this is not restrictive. An attachment member other than the hold-down **8** may be attached to the case **7**, and the elastic deformation of the engagement parts **22** in the direction of causing the receptacle **4** to fall out of the accommodation part **21** may be restricted by the attachment member.

(163) It will be understood that the present invention is not limited to the foregoing embodiments in the types, layout, and numbers of members etc., and appropriate modifications can be made with departing from the gist of the invention, like replacing the components with ones having equivalent operation and effects as appropriate.

(164) Specifically, in the foregoing first to third embodiments, the receptacle **4** is retained in the accommodation part **21**. However, this is not restrictive, and a connector having a different configuration from that of the receptacle **4** may be retained in the accommodation part **21**.

(165) In the foregoing first to third embodiments, the receptacle **4** retained in the accommodation part **21** can move up and to the left and right. However, this is not restrictive. The receptacle **4** retained in the accommodation part **21** may be movable to the front and back in addition to up and to the left and right. In such a case, the receptacle **4** retained in the accommodation part **21** can be self-aligned inside the case **2**, **5**, or **7**.

(166) In the foregoing first to third embodiments, the restriction parts **311** are disposed above the leg parts **34**. However, this is not restrictive. The restriction parts **311** may be located near the leg parts **34** other than above the leg parts **34**, like between a pair of leg parts **34**. Even in such a case, the laterally outward elastic deformation of the engagement parts **22** can be reliably restricted by the restriction parts **331**.

INDUSTRIAL APPLICABILITY

(167) The present invention is suitable for an electrical connector that includes a case configured to be mounted on a mounting surface of a substrate, and a receptacle.

REFERENCE SIGNS LIST

(168) **1** Electrical connector **2** Case **3** Hold-down **4** Receptacle **5** Case **6** Hold-down **7** Case **8** Hold-down **9** Slider **11** Electrical connector **12** Electrical connector **18** Pressing part **19** Front wall part **20** Projecting wall part **21** Accommodation part **22** Engagement part **23** Contact part **24** Mating part **25** Insertion part **26** Projecting part **27** Flat part **28** Rear wall part **29** Inner top wall part **30** Opening **31** Top surface part **32** Rear surface part **33** Internal side surface part **34** Leg part **35** Front surface part **36** External side surface part **41** Shield case **42** Contact **43** Cover **44** Housing **51** Spring part **52** Movable part **53** Rear wall part **61** Top surface part **62** Window hole part **71** Communication hole **72** Flat part **81** Top surface part **82** Window hole part **100** Substrate **100a** Mounting surface **100b** Through hole **200** Suction nozzle **221** Locking claw part **331** Restriction part **411** Mating part **412** Projecting part **421** Connection part **422** Terminal part **431** Leg part **432** Part to be engaged

Claims

1. An electrical connector to be mounted on a mounting surface of a substrate, the electrical connector comprising: a case; a receptacle retained in the case; and a slider retained in the case, the case including an accommodation part that is disposed in a bottom surface to be opposed to the mounting surface and is configured to accommodate the receptacle so that the receptacle is exposed outside, a retaining part configured to retain the receptacle in the accommodation part, a contact part configured to make contact with the mounting surface, and a communication hole that is extended upward from the accommodation part and open to outside at an upper end thereof, the slider being vertically movably retained in the communication hole, the receptacle including a terminal part to be soldered to the mounting surface, the terminal part being located at a same height as the contact part or below the contact part and being movable upward in a state where the receptacle is retained in the accommodation part by the retaining part, the slider being capable of moving down in the communication hole to press the receptacle retained in the accommodation part by the retaining part down.

2. The electrical connector according to claim 1, wherein: the case includes a projecting part configured to project from the substrate when the electrical connector is mounted on the mounting surface; and the electrical connector has a barycentric position at a portion overlapping the substrate when mounted on the mounting surface and seen from above.

3. A method for manufacturing an electrical connector to be mounted on a mounting surface of a substrate, the method comprising: an insertion step of inserting a slider into a communication hole and inserting a receptacle into an accommodation part, the accommodation part being disposed in a bottom surface of a case to be opposed to the mounting surface, the communication hole being extended upward from the accommodation part and open to outside at an upper end thereof; and a

retaining step of retaining the receptacle inserted into the accommodation part by the insertion step in the accommodation part by a retaining part of the case in a state of being exposed outside so that a terminal part of the receptacle to be soldered to the mounting surface is located at a same height as a contact part of the case to make contact with the mounting surface or below the contact part and the receptacle is movable upward, and retaining the slider inserted into the communication hole by the insertion step in the communication hole in a state of being vertically movable so that the slider moving down in the communication hole is capable of pressing the receptacle retained in the accommodation part by the retaining part down.

4. The method for manufacturing an electrical connector according to claim 3, wherein: the case includes a projecting part configured to project from the substrate when the electrical connector is mounted on the mounting surface; and the electrical connector has a barycentric position at a portion overlapping the substrate when mounted on the mounting surface and seen from above.

5. A method for mounting an electrical connector to be mounted on a mounting surface of a substrate, the method comprising, in a state where a receptacle is retained in an accommodation part disposed in a bottom surface of a case to be opposed to the mounting surface by a retaining part of the case in a state of being exposed outside so that a terminal part of the receptacle to be soldered to the mounting surface is located at a same height as a contact part of the case to make contact with the mounting surface or below the contact part and the receptacle is movable upward, and a slider is vertically movably retained in a communication hole that is extended upward from the accommodation part of the case and open to outside at an upper end thereof, transporting the electrical connector to above the mounting surface with the slider sucked to a suction nozzle and bringing the contact part into contact with the mounting surface, pressing the slider down with the suction nozzle to move the slider down so that the slider presses the receptacle down to press the terminal part against the mounting surface, and then soldering the terminal part to the mounting surface.

6. The method for mounting the electrical connector according to claim 5, wherein: the case includes a projecting part configured to project from the substrate when the electrical connector is mounted on the mounting surface; and the electrical connector has a barycentric position at a portion overlapping the substrate when mounted on the mounting surface and seen from above.
